NumPy in Python (Detailed Guide)

What is NumPy?

NumPy (Numerical Python) is a Python library used for:

- Mathematical calculations
- Working with arrays
- Data Science
- Machine Learning
- Scientific Computing
- Data Analysis

Install NumPy

```
pip install numpy
```

Import NumPy

```
import numpy as np
```

Why NumPy?

Python List:

```
numbers = [1, 2, 3, 4, 5]
```

NumPy Array:

```
import numpy as np

numbers = np.array([1, 2, 3, 4, 5])

print(numbers)
```

Advantages

- Faster than Python lists
- Less memory usage
- Supports mathematical operations
- Useful for Data Science and ML

Creating Arrays

1. 1D Array

```
import numpy as np

arr = np.array([10, 20, 30, 40])

print(arr)
```

Output:

```
[10 20 30 40]
```

---

2. 2D Array

```
arr = np.array([
    [1, 2, 3],
    [4, 5, 6]
])

print(arr)
```

Output:

```
[[1 2 3]
 [4 5 6]]
```

---

3. 3D Array

```
arr = np.array([
    [[1,2],[3,4]],
    [[5,6],[7,8]]
])

print(arr)
```

Array Attributes

```
arr = np.array([[1,2,3],[4,5,6]])
```

Shape

```
print(arr.shape)
```

Output:

```
(2, 3)
```

2 rows, 3 columns

Dimension

```
print(arr.ndim)
```

Output:

```
2
```

Size

```
print(arr.size)
```

Output:

```
6
```

Data Type

```
print(arr.dtype)
```

Output:

```
int64
```

Special Arrays

Zeros Array

```
arr = np.zeros((3,3))
```

```
print(arr)
```

Output:

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
```

Ones Array

```
arr = np.ones((2,2))
```

```
print(arr)
```

Output:

```
[[1. 1.]
 [1. 1.]]
```

Full Array

```
arr = np.full((2,3), 5)
```

```
print(arr)
```

Output:

```
[[5 5 5]
 [5 5 5]]
```

Identity Matrix

```
arr = np.eye(3)
```

```
print(arr)
```

Output:

```
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

Range Functions

arange()

```
arr = np.arange(1, 11)
```

```
print(arr)
```

Output:

```
[1 2 3 4 5 6 7 8 9 10]
```

---

linspace()

```
arr = np.linspace(1, 10, 5)
```

```
print(arr)
```

Output:

```
[ 1.    3.25  5.5   7.75 10.  ]
```

Reshaping Arrays

```
arr = np.arange(1, 13)

new_arr = arr.reshape(3,4)

print(new_arr)
```

Output:

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
```

Indexing

```
arr = np.array([10,20,30,40])

print(arr[0])
print(arr[2])
```

Output:

```
10
30
```

2D Indexing

```
arr = np.array([
    [1,2,3],
    [4,5,6]
])

print(arr[1,2])
```

Output:

```
6
```

Slicing

```
arr = np.array([10,20,30,40,50])
```

```
print(arr[1:4])
```

Output:

```
[20 30 40]
```

Mathematical Operations

```
a = np.array([1,2,3])
b = np.array([4,5,6])
```

Addition

```
print(a + b)
```

Output:

```
[5 7 9]
```

Subtraction

```
print(a - b)
```

Output:

```
[-3 -3 -3]
```

Multiplication

```
print(a * b)
```

Output:

```
[ 4 10 18]
```

Division

```
print(a / b)
```

Output:

```
[0.25 0.4  0.5 ]
```

Aggregate Functions

```
arr = np.array([10,20,30,40,50])
```

Sum

```
print(np.sum(arr))
```

Output:

```
150
```

Mean

```
print(np.mean(arr))
```

Output:

```
30.0
```

Max

```
print(np.max(arr))
```

Output:

```
50
```

Min

```
print(np.min(arr))
```

Output:

```
10
```

Standard Deviation

```
print(np.std(arr))
```

Random Numbers

Random Integer

```
arr = np.random.randint(1, 100, 5)

print(arr)
```

Random Float

```
arr = np.random.rand(5)

print(arr)
```

Sorting

```
arr = np.array([40,10,30,20])

print(np.sort(arr))
```

Output:

```
[10 20 30 40]
```

Copy vs View

Copy

```
arr = np.array([1,2,3])

new_arr = arr.copy()

new_arr[0] = 100

print(arr)
print(new_arr)
```

Output:

```
[1 2 3]
[100 2 3]
```

View

```
arr = np.array([1,2,3])

new_arr = arr.view()
```

```
new_arr[0] = 100

print(arr)
```

Output:

```
[100   2   3]
```

Boolean Filtering

```
arr = np.array([10,20,30,40,50])

print(arr[arr > 25])
```

Output:

```
[30 40 50]
```

Useful Functions

```
np.sqrt(arr)      # Square root
np.power(arr,2)   # Power
np.abs(arr)       # Absolute value
np.sin(arr)       # Sin
np.cos(arr)       # Cos
np.log(arr)       # Log
```

NumPy Example Project

Student Marks Analysis

```
import numpy as np

marks = np.array([70, 85, 90, 60, 75])

print("Total:", np.sum(marks))
print("Average:", np.mean(marks))
print("Highest:", np.max(marks))
print("Lowest:", np.min(marks))
```

Output:

```
Total: 380
Average: 76.0
Highest: 90
Lowest: 60
```

Important Interview Questions

1. What is NumPy?
2. Difference between List and NumPy Array?
3. What is ndarray?
4. What is broadcasting?
5. Difference between copy() and view()?
6. What is reshape()?
7. Difference between arange() and linspace()?
8. What is vectorization?
9. What is indexing and slicing?
10. What are shape, size, and ndim?

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